

How Many In-Context Examples Is a Task Description Worth?

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Abstract

While prompts in modern language models typically consist of both a task description and in-context examples, theoretical frameworks for in-context learning (ICL) predominantly focus on the latter. In this work, we adopt a Bayesian perspective on ICL to formally quantify the value of task descriptions. The pretraining distribution is the prior over tasks; a description and the examples are both data about the task, the description a direct but imperfect observation of it and the examples indirect observations through inputs. We measure the description’s value by its effective sample size (ESS): the number of examples that carry the same information. We analyze this in the setting of in-context linear regression, where Bayes-optimal predictions can be computed exactly. We define the ESS of a description by matching the one-step log-loss regret of a Bayes-optimal predictor using only the description to one using only examples. Our analysis reveals that a description’s worth is governed by two factors: its precision (how much it reduces prior variance) and its reliability (the probability it is correct). We show that for a reliable description, removing a fraction $1 - r$ of the prior variance in a d -dimensional task yields an ESS of approximately $d(1 - r)$. For unreliable descriptions the ESS saturates: unreliability imposes a regret floor, so beyond a point more precision buys little: at $p = 0.9$, a fiftyfold increase in precision adds a few examples where a reliable description gains a hundred. The cap lifts once examples can check the description: with n examples in hand the floor falls with n , and a precise unreliable description gains worth as examples verify it, approaching $p/(1 - p)$ times the examples in hand. Finally, small meta-trained Transformers reproduce these ESS values when trained on the matching distribution, and a Transformer trained only on reliable descriptions treats an unreliable one as a Bayes-optimal predictor with full trust would: a precise description becomes worth almost nothing. Our claims concern Bayes-optimal predictors and small meta-trained sequence models, not large language models.

1 Introduction

In-context learning (ICL) allows models to adapt to new tasks from a prompt (Brown et al., 2020). In practice, a prompt generally comprises a natural language task description paired with a few input-output demonstrations. However, formal theories of ICL largely abstract prompts as mere collections of demonstrations (Garg et al., 2022; Akyürek et al., 2023; Xie et al., 2022; Genewein et al., 2025). This leaves a significant gap in our understanding, as empirical evidence on the value of task descriptions remains mixed. Some studies show that a good description can be worth hundreds of data points (Le Scao and Rush, 2021; Honda et al., 2025), whereas others find that misleading descriptions perform about as well as informative ones (Webson and Pavlick, 2022) and that enough examples override an explicitly stated bias (Gupta et al., 2025). Even recent theoretical models that incorporate descriptors (Huang and Ge, 2025; Lin et al., 2025; Tong et al., 2026) fail to express the utility of these descriptions on a commensurate scale with the examples themselves.

We quantify the value of a task description as the number of in-context examples it can replace. Following the Bayesian interpretation of ICL (Xie et al., 2022), a sequence model can be seen as performing posterior predictive inference over a latent task. In this view the pretraining distribution is the prior over tasks, and everything in the prompt is data: the examples observe the task indirectly through inputs, and the description observes it directly, if imperfectly. Conditioning on the description first gives the prior under which the examples are processed, which is where the usual phrase “the description is the prior” comes from. Bayesian statistics measures the weight of such conditioning information by its *effective sample size* (ESS) (Morita et al., 2008), driven by two things: how concentrated the information is, and whether it agrees with the truth (Evans and Moshonov, 2006). Translating this to task descriptions, we identify two key characteristics: **precision** (the extent to which the description reduces posterior uncertainty) and **reliability** (the probability that the description is actually correct).

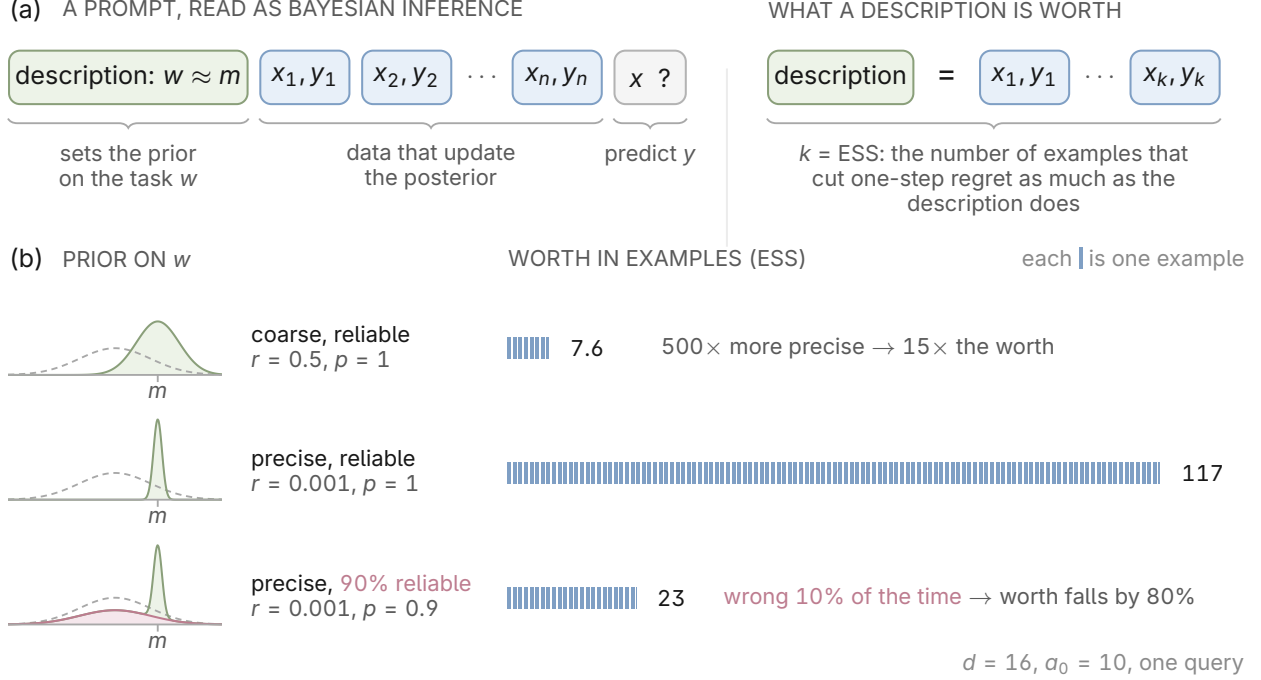


Figure 1: **A task description is worth a number of in-context examples, set by its precision and capped by its reliability.** (a) We read a prompt as Bayesian inference over a latent task w under the pretraining prior: the description is a direct noisy observation of w and the examples are indirect ones. The description’s effective sample size (ESS) is the number of examples k that lowers a Bayes-optimal predictor’s one-step regret as much as the description does. (b) Three descriptions and their ESS. Precision $1/r$ sets how narrowly the prior concentrates around the stated m ; reliability p is the chance the description is correct, and otherwise w follows the base prior (dashed; its share $1 - p$ shaded pink). Prior shapes are sketches; ESS values are from Table 2 ($d = 16, a_0 = 10$).

We compute the ESS as a function of precision and reliability within in-context linear regression (Garg et al., 2022), a setting where the Bayes-optimal predictor is exact. We focus on one-step prediction, mirroring standard prompt usage.

Contributions.

1. We introduce a formal definition for the ESS of a task description by equating the one-step log-loss regret of a description to that of n examples (§3).
2. We establish that for a perfectly reliable description, the ESS scales linearly as $\text{ESS} \approx d(1 - r)$, where d is the dimensionality and r is the precision ratio (§4).
3. We demonstrate that unreliable descriptions induce a hard floor on regret, governed by the description’s error probability $1 - p$ and its entropy $H(p)$. Consequently, the ESS saturates; making an unreliable description infinitely precise yields diminishing returns. The cap lifts as examples verify the description: its worth with n examples in hand rises with n , where a reliable description’s does not (§5).
4. We meta-train small Transformers on this process and show that they reproduce the Bayes-optimal ESS when trained on the matching distribution, and that a model trained on reliable descriptions does not discount an unreliable one (§6).

2 Related Work

Bayesian Perspectives on ICL. ICL can be framed as implicit Bayesian inference (Xie et al., 2022). In the context of linear regression, meta-trained Transformers effectively implement Bayes-optimal ridge regression (Akyürek et al.,

2023; Raventós et al., 2023; Panwar et al., 2024). Genewein et al. (2025) evaluate such models using log-loss regret against the Bayes-optimal baseline, a metric we similarly adopt. Yet, these works do not translate the influence of a prior into an equivalent number of in-context examples.

Task Descriptions in ICL Theory. Recent studies have begun exploring task descriptions theoretically. Huang and Ge (2025) include a descriptor token representing the input mean, but because it is independent of the regression weights, its ESS in our framework would be zero. Lin et al. (2025) guide models using hypothesis classes, and Tong et al. (2026) demonstrate how models can exponentially override incorrect instructions given enough examples. Lin and Lee (2024) analyze Gaussian-mixture priors similar to ours, but none of these works quantify the description’s value in terms of sample size.

Effective Sample Size. In statistics, Morita et al. (2008) define ESS via curvature matching, while Reimherr et al. (2021) highlight the need for prediction-based ESS measures. Our approach directly addresses this by defining ESS through predictive regret. Furthermore, our model of reliability draws on robust mixture priors (Schmidli et al., 2014), commonly used in clinical trials to incorporate potentially conflicting historical data.

3 Problem Formulation

Task Distribution. We consider tasks defined by a weight vector $w \in \mathbb{R}^d$. An in-context example consists of an input $x \sim \mathcal{N}(0, I_d)$ and a corresponding label $y = w^\top x + \epsilon$, where the noise is $\epsilon \sim \mathcal{N}(0, \sigma_y^2)$. The base prior over tasks is $w \sim \mathcal{N}(0, s_0^2 I_d)$. For simplicity, we set $\sigma_y = 1$ and define the prior signal-to-noise ratio (SNR) as $a_0 = s_0^2 / \sigma_y^2$.

Task Description Model. We model a task description as a vector $m \in \mathbb{R}^d$ that provides information about w . The description’s *precision ratio* is $r \in (0, 1)$, meaning the variance around the description is $s_\ell^2 = r s_0^2$. The description is generated hierarchically via $m \sim \mathcal{N}(0, (1 - r)s_0^2 I_d)$, ensuring the marginal distribution of w matches the base prior. A description is not always correct. We define its *reliability* p as the probability that the description is accurate. If correct, $w \sim \mathcal{N}(m, s_\ell^2 I_d)$; if incorrect (with probability $1 - p$), w simply follows the base prior $\mathcal{N}(0, s_0^2 I_d)$ independently of m . Consequently, the Bayes-optimal prior conditioned on the description is a robust mixture (Schmidli et al., 2014): $p \mathcal{N}(m, s_\ell^2 I_d) + (1 - p) \mathcal{N}(0, s_0^2 I_d)$. Equivalently, m is a noisy observation of the weights themselves, $m | w \sim \mathcal{N}((1 - r)w, r(1 - r)s_0^2 I_d)$ when correct, so the description is data about w and the ESS is an exchange rate between direct observations of w and indirect ones through x . A description with precision ratio r locates each weight to within $\sqrt{r}$ of its typical size s_0 : 71% at $r = 0.5$, 22% at $r = 0.05$, 3% at $r = 0.001$. The formulation treats the description as a numerical observation rather than text to be parsed.

Regret and Effective Sample Size. We evaluate predictors based on one-step predictive log-loss for a new query x . The regret is the expected difference in log-loss between the predictor and an oracle that knows w exactly, the expectation taken over tasks, inputs, and labels (Genewein et al., 2025).

Definition 1 (Effective Sample Size). *Let R^{desc} be the regret of a Bayes-optimal predictor given only the task description, and $R^{\text{ex}}(n)$ be the regret of a Bayes-optimal predictor under the base prior given n examples. The effective sample size (ESS) of the description is the value n (interpolated continuously) where $R^{\text{ex}}(n) = R^{\text{desc}}$.*

The two sides of the definition are prompts of two kinds: a prompt holding the description and nothing else, and prompts holding n examples and no description. The ESS asks what the description is worth on its own, in the currency of examples on their own. Prompts holding both are not part of the definition; they enter in §6, where we also report regret with a description and n examples together. Since we compare Bayes-optimal predictors under different conditioning sets, the ESS measures information content rather than the specific architecture of a learner.

Definition 2 (Worth with examples in hand). *Let $R^{\text{desc}}(n)$ be the regret of the Bayes-optimal predictor given the description and n examples. The worth of the description with n examples in hand is $\text{ESS}_n = n^* - n$, where n^* (interpolated continuously) solves $R^{\text{ex}}(n^*) = R^{\text{desc}}(n)$: the further examples an examples-only predictor needs to match it. $\text{ESS}_0 = \text{ESS}$.*

Table 1: Exact ESS of a reliable description ($p = 1$) for $a_0 = 10$, averaged over three seeds (standard deviation across seeds below 2% of each value). The approximation $d(1 - r)$ holds well as long as the exact ESS does not exceed d .

d	r	ESS	$d(1 - r)$
4	0.5	1.77	2.0
4	0.2	3.32	3.2
16	0.5	7.62	8.0
16	0.2	12.6	12.8
16	0.05	16.9	15.2
16	0.01	26.1	15.8
64	0.5	31.6	32.0
64	0.2	50.7	51.2
64	0.05	62.1	60.8
64	0.01	73.5	63.4

4 The Value of Precision

We first analyze a perfectly reliable description ($p = 1$). Given n examples $X_n \in \mathbb{R}^{n \times d}$, the posterior covariance under a Gaussian prior with SNR a is $\Sigma = \sigma_y^2(I/a + X_n^\top X_n)^{-1}$. This is independent of the examples' labels and the description m . The expected one-step regret simplifies to $\frac{1}{2} \mathbb{E} \log(1 + x^\top \Sigma x)$. A reliable description equates to $n = 0$ and $a = ra_0$, whereas relying solely on examples uses $a = a_0$.

Proposition 1. *Let ψ be the digamma function. As $a_0 \rightarrow \infty$, for $n < d$,*

$$R^{\text{ex}}(n) = \frac{1}{2} \log a_0 + \frac{1}{2} \log 2 + \frac{1}{2} \psi\left(\frac{d-n}{2}\right) + o(1), \quad R^{\text{desc}} = \frac{1}{2} \log(ra_0) + \frac{1}{2} \log 2 + \frac{1}{2} \psi\left(\frac{d}{2}\right) + o(1),$$

so the high-SNR ESS, with $R^{\text{ex}}(n)$ extended to non-integer n through the digamma function, solves $\psi\left(\frac{d-n}{2}\right) - \psi\left(\frac{d}{2}\right) = \log r$. As $rd \rightarrow \infty$, uniformly in $r \in (0, 1)$, its solution is

$$\text{ESS} = (d - 1)(1 - r) + O\left(\frac{1}{rd}\right).$$

Idea. Regret is half the log of the leftover variance in the prediction. At high SNR each example removes all the variance along the one direction it looks at, so $n < d$ examples leave $d - n$ directions untouched; a description shrinks all d directions by the factor r . Matching the two leftovers gives $d - n \approx rd$. The proof, with the error term, is in Appendix A.

The error term makes the range of validity precise: the approximation needs $rd \gg 1$, that is, the description must leave many more than one example's worth of variance unexplained, which is the condition $\text{ESS} < d$ in the first-order form $\text{ESS} \approx d(1 - r)$. At $a_0 = 100$ the exact ESS is within 0.1 of $(d - 1)(1 - r)$ for $d \in \{16, 64\}$ and $r \in \{0.5, 0.2\}$ (values 7.50, 12.1, 31.6, 50.5 against 7.5, 12.0, 31.5, 50.4).

Table 1 illustrates this alignment. A description eliminating 50% of the prior variance is worth roughly half the dimension in examples. However, once the description is precise enough that its ESS surpasses d , the approximation breaks down. The true ESS scales beyond d because d noisy examples cannot pinpoint w perfectly.

5 The Bottleneck of Reliability

In reality, descriptions can be flawed or ambiguous. When providing an unreliable description without examples, the Bayes-optimal predictive distribution becomes a mixture. Without observations that could identify the correct component, the predictor must average over both.

Proposition 2. *Let $R^{\text{desc}}(p, r)$ be the regret of a description with reliability p and precision ratio r , $R_{p=1}^{\text{desc}}(r)$ the same for a reliable one, and $R^{\text{ex}}(0)$ the regret with neither description nor examples. For every $p \in (0, 1)$ and $r \in (0, 1)$,*

$$p R_{p=1}^{\text{desc}}(r) + (1 - p) R^{\text{ex}}(0) \leq R^{\text{desc}}(p, r) \leq p R_{p=1}^{\text{desc}}(r) + (1 - p) R^{\text{ex}}(0) + H(p),$$

where $H(p)$ is the binary entropy in nats. In particular $R^{\text{desc}}(p, r) \geq (1 - p) R^{\text{ex}}(0)$ for every r , and as $r \rightarrow 0$ the regret lies between $(1 - p) R^{\text{ex}}(0)$ and $(1 - p) R^{\text{ex}}(0) + H(p)$.

Table 2: ESS of unreliable descriptions ($d = 16$, $a_0 = 10$), three seeds; standard deviation across seeds at most 2% of each value. As precision increases (smaller r), the ESS is capped by the reliability p .

r	$p = 1$	$p = 0.99$	$p = 0.9$	$p = 0.7$
0.5	7.6	7.5	6.3	4.1
0.05	16.9	16.6	14.8	11.7
0.001	116.5	64.2	23.1	15.6

Idea. Compare the learner with a twin who is also told whether the description is correct. The twin’s regret is $p R_{p=1}^{\text{desc}}(r) + (1 - p) R^{\text{ex}}(0)$. The learner cannot beat the twin, since extra information never hurts a Bayes-optimal predictor. And it loses at most $\log(1/p)$ nats when the description is correct and $\log(1/(1 - p))$ when it is wrong, because its predictive puts those weights on the twin’s answer; the average is $H(p)$. The proof is in Appendix B.

The gap between the two bounds is the cost of not knowing whether the description is correct, and it is at most $H(p)$ however precise the description. For $d = 16$ and $a_0 = 10$, $R^{\text{ex}}(0) = 2.51$ nats and $R_{p=1}^{\text{desc}}(0.001) = 0.07$; at $p = 0.9$ the two bounds are 0.32 and 0.64, and the computed regret at $r = 0.001$ is 0.57.

Because $R^{\text{ex}}(n)$ decreases with n , the lower bound caps the ESS at the n where $R^{\text{ex}}(n) = (1 - p) R^{\text{ex}}(0)$, however precise the description. An unreliable description can replace only a bounded number of examples, and the identification cost, up to $H(p)$, lowers the cap further.

Table 2 demonstrates this saturation. For a perfectly reliable description, refining r from 0.05 to 0.001 raises the ESS from 16.9 to 116.5. However, if $p = 0.9$, the ESS only grows from 14.8 to 23.1. Even a 1% unreliability ($p = 0.99$) nearly halves the maximum potential ESS of the most precise description. The same saturation patterns manifest at $d \in \{4, 64\}$ and $a_0 = 100$.

Examples lift the bottleneck. With no examples there is nothing to check a description against, so hedging by $1 - p$ is forced. Examples change this: they can reveal whether the description is correct, and once they have, the predictor can use its full precision. The floor of Proposition 2 generalises to n examples in hand, and it falls with n .

Proposition 3. Let $R_{p=1}^{\text{desc}}(n)$ be the regret with a reliable description and n examples, and π_n the Bayes-optimal predictor’s posterior probability that the description is correct after n examples. For every n , p , and r ,

$$p R_{p=1}^{\text{desc}}(n) + (1 - p) R^{\text{ex}}(n) \leq R^{\text{desc}}(n) \leq p R_{p=1}^{\text{desc}}(n) + (1 - p) R^{\text{ex}}(n) + \mathbb{E}[-\log \pi_n(Z)],$$

where $\pi_n(Z)$ is the posterior weight on the true component. The identification term $\mathbb{E}[-\log \pi_n(Z)]$ equals $H(p)$ at $n = 0$ and is non-increasing in n .

Idea. The twin argument of Proposition 2 goes through with the examples included: the twin who is told whether the description is correct has regret $p R_{p=1}^{\text{desc}}(n) + (1 - p) R^{\text{ex}}(n)$, the learner cannot beat it, and the learner’s predictive puts weight π_n on the twin’s answer, so it loses at most $-\log \pi_n(Z)$. More examples can only sharpen the posterior on Z in expectation. The proof is in Appendix C.

Two consequences follow. As $r \rightarrow 0$ the floor is $(1 - p) R^{\text{ex}}(n)$, which decreases with n , so the cap on the description’s worth rises as examples accumulate. And since $R^{\text{ex}}(n) \approx d/(2n)$ for large n , matching $(1 - p) d/(2n)$ needs $n^* \approx n/(1 - p)$ examples, so the worth of a very precise unreliable description grows like $\text{ESS}_n \approx n p/(1 - p)$: nine times the examples in hand at $p = 0.9$. A reliable description, by contrast, is worth about $1/(r a_0)$ examples at large n , a constant. Figure 2 shows both for the exact learner: with $r = 0.001$ and $p = 0.9$ at $d = 16$, the description is worth 23 examples alone, 30 after one example, 32 after three, and 57 after twenty-four, while the reliable description stays near 115. The curves are far from their limits at these n , but the direction is fixed by the proposition: reliability caps precision until examples can check the description.

6 Meta-Trained Transformers

Design. To test whether a learned predictor reproduces these values, we meta-train Transformers on prompts generated by the process of §3 and compare them with the Bayes-optimal predictor on the same prompts, following Genewein et al. (2025). Prompts use the prefix embedding of Huang and Ge (2025): an optional descriptor token carrying m , up

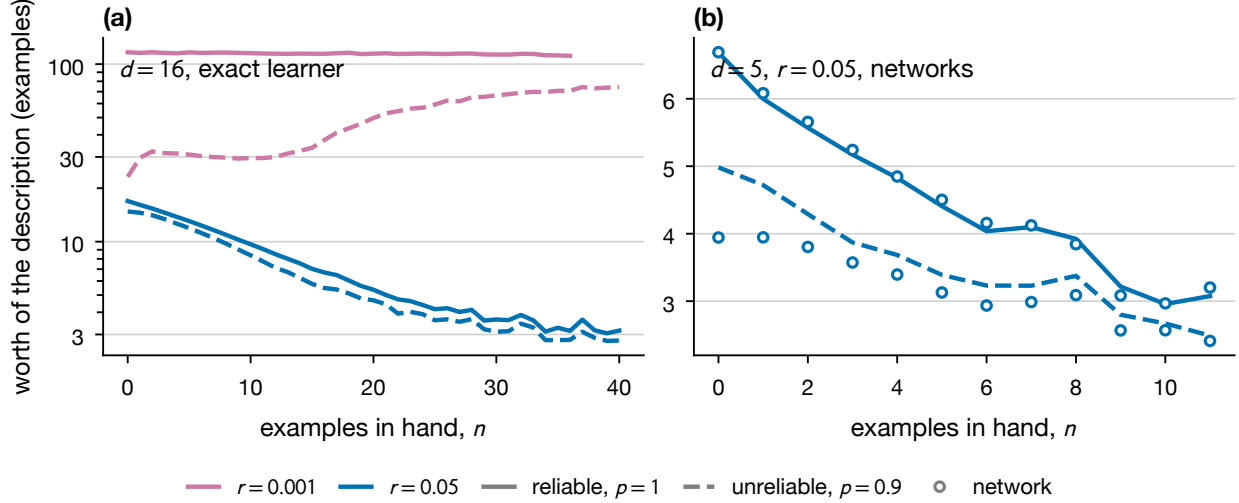


Figure 2: **An unreliable description gains worth as examples verify it; a reliable one does not.** Worth of the description with n examples in hand (Definition 2), as further examples saved. (a) Exact learner, $d = 16$, $a_0 = 10$, mean of three seeds of 2,000 prompts; log scale. (b) Trained networks at $d = 5$, $r = 0.05$ (markers, one model per reliability) against the exact learner (lines). The network trained at $p = 0.9$ under-values its description alone and converges to the exact learner as examples arrive.

to 15 example tokens each holding an input beside its answer, and a final query token, with two indicator coordinates marking the token type. There is no positional encoding, so the examples are exchangeable. The architecture and optimisation follow Garg et al. (2022): 12 layers, 8 heads, width 256, Adam at 10^{-4} , fresh prompts at every step. The network outputs a mean and a log variance for the answer to the query and is trained on the Gaussian log loss, so its regret is on the scale of $\$3$. We set $d = 5$ and $a_0 = 10$, draw the number of examples uniformly from 0 to 15, and omit the descriptor in half the prompts. The descriptor supplies only m : each network is trained at one precision r and one reliability p , which it never sees and must learn from the training distribution. Every model trains for 40,000 steps at batch 1024, one model per setting. A pilot confirmed the budget: the mean gap between the network and the Bayes-optimal predictor falls from 0.07 nats at 5,000 steps to 0.02 at 20,000 and stays there to 40,000.

Readouts. We evaluate each network on 20,000 fresh prompts for every n from 0 to 15, with and without a descriptor. The ESS uses two of these conditions, exactly as in Definition 1: the network’s regret on prompts holding the descriptor alone is placed on its own regret curve for prompts holding n examples alone, and the crossing is the ESS. The remaining conditions, a descriptor followed by $n \geq 1$ examples, show how the network combines the two and are plotted alongside. We also evaluate the two models trained at $p = 1$ on prompts whose descriptors are correct only with probability 0.9, which asks how a learner that has never seen a wrong description treats one.

Results. Table 3 and Figure 3 give the readouts. For reliable descriptors the networks reproduce the Bayes-optimal values: ESS 6.7 against 6.7 for the precise descriptor and 2.2 against 2.3 for the coarse one, with regret within 0.02 nats of the Bayes-optimal predictor at every n . The coarse unreliable descriptor is also reproduced, 1.8 against 1.9. The precise unreliable descriptor is not: the network’s ESS is 3.9 against 5.0, its regret from the descriptor alone is 0.3 nats above the Bayes-optimal predictor’s, and the trust that best explains its predictions is 0.85 rather than 0.9. The gap closes as examples arrive (Figure 3c). This is not under-training: continuing the same model to 80,000 steps on fresh prompts leaves its ESS at 3.9 and its implied trust at 0.85. With one trained model per setting we report it as an observation rather than a finding.

Transfer to unreliable descriptors. The models trained at $p = 1$ behave like a Bayes-optimal predictor that trusts the descriptor fully. On prompts where one descriptor in ten is wrong, the model trained on precise reliable descriptors gets almost nothing from a descriptor: ESS 0.7, where a calibrated predictor would get 5.0, and the fully trusting

Table 3: Networks against the Bayes-optimal predictor ($d = 5$, $a_0 = 10$, one trained model per row, 20,000 evaluation prompts). “Trained” is the Bayes-optimal predictor with trust equal to the training reliability p_{train} , the predictor each network is trained toward; “calibrated” is the predictor with trust equal to the test reliability, shown where the two differ. The last two columns are the mean regret gap, network minus Bayes-optimal, over $n = 0, \dots, 15$, with and without a descriptor.

r	p_{train}	p_{test}	ESS			Regret gap (nats)	
			Network	Trained	Calibrated	Descriptor	None
0.05	1	1	6.7	6.7	—	0.023	0.022
0.5	1	1	2.2	2.3	—	0.019	0.019
0.05	0.9	0.9	3.9	5.0	—	0.070	0.025
0.5	0.9	0.9	1.8	1.9	—	0.019	0.016
0.05	1	0.9	0.7	0.0	5.0	0.000	0.022
0.5	1	0.9	1.6	1.7	1.9	0.024	0.019

Bayes-optimal predictor gets 0. Its implied trust is 0.99. The coarse model is forgiven its trust, 1.6 against the calibrated 1.9, because a coarse descriptor that is wrong does little harm. This is the mechanism of §5 seen in a trained learner: the cost of unreliability falls on precise descriptors, and a learner that has never seen a wrong descriptor pays it in full.

7 Discussion

Our analysis offers a precise vocabulary for discussing the utility of task descriptions in prompts. For a Bayes-optimal predictor, a task description is worth a specific number of in-context examples. Assuming perfect reliability, this worth scales linearly with the proportion of prior variance the description eliminates. However, reliability acts as a strict bottleneck. Because a predictor must hedge against the possibility that the description is incorrect, the value of very precise descriptions saturates if they are not perfectly reliable. The hedge is only forced while nothing can check the description: examples lift it, and an unreliable precise description is worth more after a few examples than alone. Reliability, then, is not a fixed discount on a description but a cost that examples pay down.

Limitations and Future Work. Our theoretical claims are restricted to Bayes-optimal predictors and small-scale meta-trained Transformers in a linear-Gaussian setting, chosen to permit exact calculation. Scaling these insights to large language models operating on complex text remains an open challenge. Additionally, our ESS metric applies to one-step prediction. Over long horizons, sequential feedback allows predictors to detect and adapt to incorrect descriptions, meaning ESS will dynamically evolve with the prediction horizon—a compelling avenue for future research.

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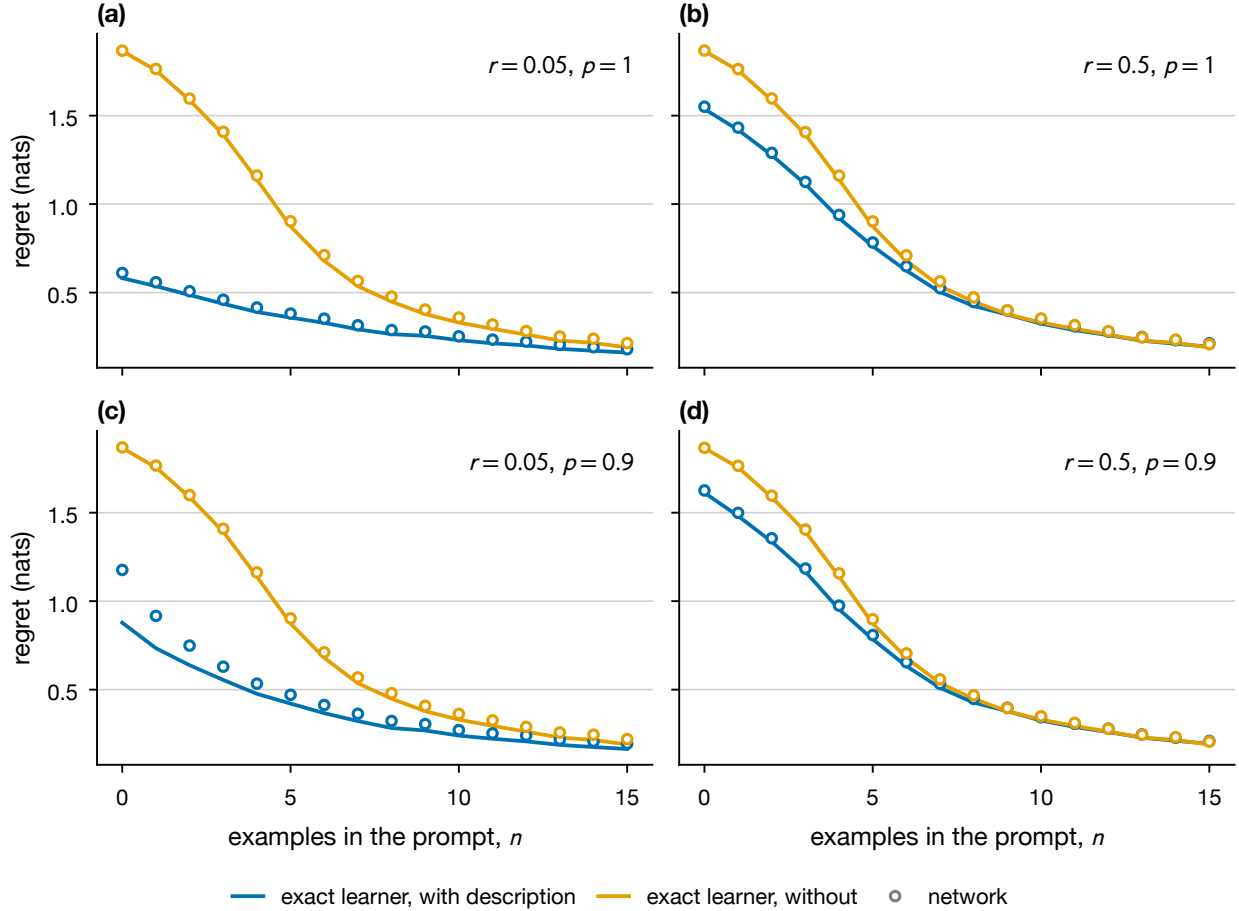


Figure 3: **Networks trained on the matching distribution track the Bayes-optimal predictor, except the precise unreliable descriptor.** Regret against the number of examples for the network (markers, standard errors over 20,000 prompts) and the Bayes-optimal predictor (lines); one trained model per panel. The lower curve in each panel is prompts with a descriptor followed by n examples, the upper curve prompts with n examples alone. The ESS is the n at which the upper curve reaches the lower curve’s value at $n = 0$. (a) $r = 0.05, p = 1$; (b) $r = 0.5, p = 1$; (c) $r = 0.05, p = 0.9$; (d) $r = 0.5, p = 0.9$. In (c) the network’s regret with the descriptor alone exceeds the Bayes-optimal predictor’s by 0.3 nats, and its ESS is 3.9 against 5.0.

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A Proof of Proposition 1

Idea. Regret is half the log of the leftover variance in the prediction. At high SNR each example removes all the variance along the one direction it looks at and none elsewhere, so $n < d$ examples leave $d - n$ directions untouched. A description instead shrinks every one of the d directions by the factor r . The ESS is the n at which the two leftovers match.

Proof. Step 1: regret is leftover variance. The Bayes predictive for y is the true conditional law of y given the data, a Gaussian with variance $1 + x^\top \Sigma x$ where Σ is the posterior covariance of w . The expected log loss of a true conditional law is its entropy, $\frac{1}{2} \log(2\pi e(1 + x^\top \Sigma x))$; the oracle’s predictive is $\mathcal{N}(w^\top x, 1)$ with entropy $\frac{1}{2} \log(2\pi e)$. So the regret is $\frac{1}{2} \mathbb{E} \log(1 + x^\top \Sigma x)$: half the log of the leftover variance.

Step 2: examples. Write $X_n^\top X_n = \sum_{i \leq n} \lambda_i u_i u_i^\top$ with $\lambda_i > 0$ almost surely, and let $P_\perp$ project onto the $d - n$ directions the examples did not look at. Then $\Sigma = (I/a_0 + X_n^\top X_n)^{-1}$ has eigenvalue a_0 on those directions and $1/(1/a_0 + \lambda_i)$ on the others, so

$$1 + x^\top \Sigma x = a_0 \|P_\perp x\|^2 + T, \quad T = 1 + \sum_{i \leq n} \frac{(u_i^\top x)^2}{1/a_0 + \lambda_i}.$$

Because x is a fresh isotropic Gaussian, $\|P_\perp x\|^2 \sim \chi_{d-n}^2$, independent of X_n . Hence

$$R^{\text{ex}}(n) = \frac{1}{2} \log a_0 + \frac{1}{2} \mathbb{E} \log \chi_{d-n}^2 + \frac{1}{2} \mathbb{E} \log \left(1 + \frac{T/a_0}{\|P_\perp x\|^2} \right).$$

The last term is nonnegative, and $T/a_0 = 1/a_0 + \sum_{i \leq n} (u_i^\top x)^2 / (1 + a_0 \lambda_i)$ decreases to 0 as $a_0 \rightarrow \infty$, so by monotone convergence the term is $o(1)$. (It is finite at any finite a_0 because $R^{\text{ex}}(n)$ is.)

Step 3: description. Here $n = 0$ and $\Sigma = r a_0 I$, so $1 + x^\top \Sigma x = 1 + r a_0 \|x\|^2$ with $\|x\|^2 \sim \chi_d^2$, and the same argument gives $R^{\text{desc}} = \frac{1}{2} \log(r a_0) + \frac{1}{2} \mathbb{E} \log \chi_d^2 + o(1)$.

Step 4: match them. The $\log a_0$ terms cancel, and $\mathbb{E} \log \chi_k^2 = \psi(k/2) + \log 2$. Taking $\psi((d-n)/2)$ as the extension of $R^{\text{ex}}(n)$ to non-integer n , the high-SNR ESS solves

$$\psi\left(\frac{d-n}{2}\right) - \psi\left(\frac{d}{2}\right) = \log r,$$

which has one root in $(0, d)$ because ψ is increasing and $\log r < 0$. Read it as: $d - n$ untouched directions must be as big, on a log scale, as d directions each shrunk by r . Replacing $\mathbb{E} \log \chi_k^2$ by $\log k$ gives the first-order answer $d - n = rd$, that is $\text{ESS} \approx d(1 - r)$.

Step 5: the error term. Let $k = d - n$. As $rd \rightarrow \infty$, $\psi(k/2) = \psi(d/2) + \log r \rightarrow \infty$ forces $k \rightarrow \infty$. The log of a χ_k^2 variable averages a little below the log of its mean: $\psi(z) = \log z - \frac{1}{2z} + O(z^{-2})$ as $z \rightarrow \infty$. Hence

$$\log \frac{k}{d} - \frac{1}{k} + \frac{1}{d} + O(k^{-2}) = \log r \implies k = rd \left(1 + \frac{1}{k} - \frac{1}{d} + O(k^{-2}) \right).$$

So $k = rd(1 + o(1))$ and $rd/k = 1 + O(1/k)$. Multiplying out, $k = rd + 1 - r + O(1/(rd))$, and $\text{ESS} = d - k = (d-1)(1-r) + O(1/(rd))$. The correction $-(1-r)$ comes from the $-1/k + 1/d$ term: the $d-n$ untouched directions are fewer, so their chi-square sits further below its mean on the log scale. $\square$

B Proof of Proposition 2

Idea. Compare the learner with a twin who is also told whether the description is correct. The twin's regret is easy to write down. The learner cannot beat the twin, because extra information never hurts a Bayes-optimal predictor. And the learner cannot be much worse: its predictive puts weight p on the twin's answer when the description is correct and $1-p$ when it is wrong, so it loses at most $\log(1/p)$ or $\log(1/(1-p))$ nats against the twin, which averages to $H(p)$.

Proof. Let $Z = 1$ if the description is correct and $Z = 0$ otherwise.

Step 1: the predictive is a p -mixture. Seeing m says nothing about Z , because m has the same law $\mathcal{N}(0, (1-r)a_0I)$ either way, and x is independent of Z . With no examples there is nothing else to update on, so the posterior weight on $Z = 1$ stays p and the Bayes predictive is

$$q(y) = p f_1(y) + (1-p) f_0(y), \quad f_1 = \mathcal{N}(m^\top x, 1 + ra_0\|x\|^2), \quad f_0 = \mathcal{N}(0, 1 + a_0\|x\|^2),$$

where f_1 is the predictive given a correct description and f_0 the predictive under the base prior.

Step 2: the twin. Given Z , the true conditional density of y is f_Z . So the twin predicts with f_1 when $Z = 1$, with regret $R_{p=1}^{\text{desc}}(r)$, and with f_0 when $Z = 0$, with regret $R^{\text{ex}}(0)$. Its regret is $p R_{p=1}^{\text{desc}}(r) + (1-p) R^{\text{ex}}(0)$.

Step 3: the learner cannot beat the twin (lower bound). Given (x, m, Z) , the expected log loss of any density q exceeds that of the true density f_Z by $\text{KL}(f_Z \| q) \geq 0$. Averaging over Z gives $R^{\text{desc}}(p, r) \geq p R_{p=1}^{\text{desc}}(r) + (1-p) R^{\text{ex}}(0)$.

Step 4: the learner is at most $H(p)$ worse (upper bound). Pointwise, $q \geq p f_1$ and $q \geq (1-p) f_0$. So when $Z = 1$, $-\log q \leq -\log f_1 + \log(1/p)$, and when $Z = 0$, $-\log q \leq -\log f_0 + \log(1/(1-p))$. Averaging, the learner's regret exceeds the twin's by at most $p \log(1/p) + (1-p) \log(1/(1-p)) = H(p)$.

Step 5: the limits. $R_{p=1}^{\text{desc}}(r) = \frac{1}{2} \mathbb{E} \log(1 + ra_0\|x\|^2)$ is nonnegative, which gives the floor $(1-p) R^{\text{ex}}(0)$ for every r , and it decreases to 0 as $r \rightarrow 0$ by monotone convergence, which gives the two limits. $\square$

C Proof of Proposition 3

Proof. Let $D_n = (x_{1:n}, y_{1:n}, m, x)$ be everything the predictor sees before predicting y , and Z the indicator that the description is correct. The Bayes-optimal predictive is $q(y) = \pi_n f_1(y) + (1 - \pi_n) f_0(y)$, where $\pi_n = P(Z = 1 \mid D_n)$ and f_1, f_0 are the predictives of the two components given D_n : f_1 is the reliable-description predictor's, and f_0 is the examples-only predictor's, because under $Z = 0$ the description is independent of w and carries no information about y .

Lower bound. Given (D_n, Z) the true conditional density of y is f_Z , and $\mathbb{E}[-\log q \mid D_n, Z] - \mathbb{E}[-\log f_Z \mid D_n, Z] = \text{KL}(f_Z \| q) \geq 0$. Averaging over Z , and noting that the regret of f_1 on $Z = 1$ is $R_{p=1}^{\text{desc}}(n)$ and the regret of f_0 on $Z = 0$ is $R^{\text{ex}}(n)$, gives the lower bound.

Upper bound. Pointwise $q \geq \pi_n f_1$ and $q \geq (1 - \pi_n) f_0$, so $-\log q \leq -\log f_Z - \log \pi_n(Z)$ with $\pi_n(1) = \pi_n$ and $\pi_n(0) = 1 - \pi_n$. Taking expectations gives the upper bound with the identification term $\mathbb{E}[-\log \pi_n(Z)]$.

The identification term. At $n = 0$ nothing updates Z (Appendix B), so $\pi_0 = p$ and $\mathbb{E}[-\log \pi_0(Z)] = H(p)$. For $n \geq 1$, $\mathbb{E}[-\log \pi_n(Z)] = H(Z \mid D_n)$, the conditional entropy of Z given the data, and conditioning on more data does not increase conditional entropy: $H(Z \mid D_{n+1}) \leq H(Z \mid D_n)$. $\square$